Overview

The goal of this roadmap is to become **very comfortable with Java syntax, Core Java, modern Java features, and eventually Java backend development with Spring Boot**.

The learning path is divided into 10 levels.

🟢 Level 1 — Java Basics & Coding Challenges

Status: ✅ Completed

Focus:

- Variables and data types
- Strings
- Loops
- Conditions
- Arrays
- Methods
- `StringBuilder`
- `HashMap`
- `HashSet`
- Basic problem solving

🟢 Level 2 — Java Collections, Comparators & Streams

Status: ✅ Completed

Focus:

- `ArrayList`
- `HashSet`
- `LinkedHashSet`
- `TreeSet`
- `HashMap`
- `LinkedHashMap`
- `Queue`
- `ArrayDeque`

- `PriorityQueue`
- Min Heap / Max Heap
- `Comparator`
- Sorting
- `groupingBy`
- Java Streams
- `Optional`
- LRU Cache concepts

🟢 Level 3 — OOP & Core Java

Status: 🟡 In Progress / Next

Focus:

- Classes & Objects
- Constructors
- `this`
- `static`
- `final`
- Encapsulation
- Getters / Setters
- Inheritance
- `super`
- Method Overloading
- Method Overriding
- Polymorphism
- Abstract Classes
- Interfaces
- Composition vs Inheritance
- Enums
- Records
- Sealed Classes

🟢 Level 4 — Exception Handling & Generics

Focus:

- `try`
- `catch`
- `finally`
- Checked Exceptions
- Unchecked Exceptions
- `throw`
- `throws`
- Custom Exceptions
- Generics
- Generic Classes
- Generic Methods
- Bounded Generics
- `? extends`
- `? super`
- Wildcards

🟢 Level 5 — Modern Java

Focus:

- Lambda Expressions
- Functional Interfaces
- `Predicate`
- `Function`
- `Consumer`
- `Supplier`
- Method References
- `Optional`
- Advanced Streams
- `map`
- `flatMap`
- `filter`
- `reduce`
- `collect`
- `groupingBy`
- `partitioningBy`

🟡 Level 6 — Multithreading & Concurrency

Focus:

- Threads
- `Runnable`
- `Callable`
- `ExecutorService`
- Thread Pools
- `Future`
- `CompletableFuture`
- `synchronized`
- Locks
- Atomic Classes
- Concurrent Collections
- Race Conditions
- Deadlocks

🟡 Level 7 — JVM & Java Internals

Focus:

- JVM
- JDK vs JRE
- Stack vs Heap
- Garbage Collection
- String Pool
- `==` vs `.equals()`
- `hashCode()`
- How `HashMap` works
- How `ArrayList` works
- How `HashSet` works
- Immutability
- `final`
- Java Pass-by-Value
- Memory Management

🔵 Level 8 — Java Backend Fundamentals

Focus:

- Maven / Gradle
- Java Project Structure
- JDBC
- PostgreSQL
- Connection Pooling
- REST APIs
- HTTP
- JSON
- Configuration
- Environment Variables
- Logging
- JUnit
- Mockito

🔵 Level 9 — Spring Boot

Focus:

- Spring Boot
- Dependency Injection
- IoC
- Controllers
- Services
- Repositories
- DTOs
- Validation
- Exception Handling
- Spring Data JPA
- Hibernate
- PostgreSQL
- Transactions
- Pagination
- Spring Security
- Authentication

- Authorization
- REST API Design

🔴 Level 10 — Production Java Backend Project

Build one complete backend project combining everything.

Example: Transaction Management API

```
Client
   ↓
REST Controller
   ↓
Service Layer
   ↓
Repository
   ↓
PostgreSQL
```

Features:

- Authentication
- Users
- Accounts
- Transactions
- Pagination
- Filtering
- Sorting
- Validation
- Exception Handling
- Database Transactions
- Unit Tests
- Integration Tests
- Docker
- API Documentation

Progress Tracker

✅ Level 1 — Java Basics & Coding Challenges

✅ Level 1 — Java Basics & Coding Challenges
✅ Level 2 — Collections & Streams
⬜ Level 3 — OOP & Core Java
⬜ Level 4 — Exceptions & Generics
⬜ Level 5 — Modern Java
⬜ Level 6 — Multithreading & Concurrency
⬜ Level 7 — JVM & Java Internals
⬜ Level 8 — Java Backend Fundamentals
⬜ Level 9 — Spring Boot
⬜ Level 10 — Production Backend Project